

NUMERICAL CLAIM BOUNDARY

Exact after representation - not exact before it

Only the fixed multiset of finite binary64 operands enters the exact-sum contract.

1. Earlier quantity

Probability, logarithm,
MI estimate, redundancy,
or weighted PID term

May already contain:
sampling, model, log,
product, or
estimator error

2. Represented inputs

Finite 64-bit payloads

$$x_i = X_i 2^{-1074}$$

Sign, exponent, and
53-bit significand decoded

EXACT CLAIM
STARTS HERE

3. Exact accumulation

34 unsigned 64-bit limbs
for each sign magnitude

$$P = \sum X_i^+$$

$$N = \sum X_i^-$$

No intermediate rounding
Fixed storage
under term bound

4. Subtract once; round once

$$D = |P - N|$$

$$y = \text{RN_even}(\text{sign}(P-N) \\ D 2^{-1074})$$

Guard + sticky + retained parity
choose the binary64 payload.

Exact cancellation gives
positive zero.

PERMITTED CLAIM

The same finite operand multiset gives one
correctly rounded result in every order.

Bounded fixtures support Rust conformance;
they are not a universal proof.

PROHIBITED INFERENCE

The PID, probability, logarithm, or estimator
is therefore exact, unbiased, or calibrated.

Upstream error and application assumptions
remain separate obligations.